SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 14

Case Study: Online Examination System – Negative Test Scenarios and Test Cases

AIM:

To identify and prepare negative test scenarios and test cases for the Online Examination System to verify that the application properly handles invalid inputs, unauthorized access, expired sessions, late submissions, and unexpected conditions while ensuring security, data integrity, and reliable system operation.

1. Problem Description

Negative testing verifies rejection of invalid credentials, unauthorized exam access, late submissions, invalid answers and attempts to submit or access an exam outside its rules.

2. Functional Requirements

1. Only registered and authorized students can access assigned exams.
2. Exam availability must follow configured dates and times.
3. Questions and answers must be saved according to system rules.
4. The timer and submission deadline must be enforced.
5. Each exam attempt must have a consistent submission status.
6. Students must not access another student's exam or result.
7. Results must be visible only when published and authorized.

3. Negative Test Cases

Negative testing verifies that invalid, incomplete, or unexpected inputs are rejected safely with suitable messages.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Login with invalid credentials	Enter incorrect username/password	Login denied	Pass
NT02	Open exam without authorization	Use an account not assigned to the exam	Access denied	Pass
NT03	Start before exam window	Attempt to start before permitted time	Exam does not start	Pass
NT04	Submit after deadline	Attempt submission after the allowed time	Late submission handled according to exam rules	Pass

NT05	Leave mandatory answer invalid	Enter an invalid response format	Validation message shown or answer rejected	Pass
NT06	Submit twice	Click Submit repeatedly	Only one final submission is recorded	Pass
NT07	Access another student's result	Change result identifier or URL	Access denied; other result not disclosed	Pass
NT08	Resume after exam closure	Try to reopen a completed/expired exam	Further answering is blocked	Pass
NT09	Use expired session	Attempt to submit after session expiry	Re-authentication or safe recovery required; no data leakage	Pass
NT10	Disconnect during exam	Simulate temporary network loss	Saved answers are retained or recovery follows documented rules	Pass

4. Detailed Execution of a Failed Test Case

Example: Select a test case that fails during actual execution. Record the precondition, steps, expected result, and actual result. Do not mark a test as failed unless the defect is observed.

5. Sample Bug Report

Field	**Details**
Bug ID	BR-001
Linked Test Case	Relevant failed negative test case
Severity / Priority	Assign according to impact and urgency
Expected Result	Invalid operation is rejected with a clear message
Actual Result	Application accepts or processes the invalid operation
Suggested Fix	Add server-side validation and repeat checks before final processing
Status	Open – awaiting developer investigation and correction

Conclusion

The negative test cases help verify that the Online Examination System meets its requirements and responds appropriately to the tested conditions.